IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received July 15, 2019, accepted July 29, 2019, date of publication August 2, 2019, date of current version August 19, 2019.

Digital Object Identifier 10.1109/ACCESS.2019.2932720

Multi-Objective Optimization for Multi-Line and U-Shape Layout of Sewing Assembly Line in Apparel Industry

XUJING ZHANG, LICHUAN WANG, AND YAN CHEN
College of Textile and Clothing Engineering, Soochow University, Suzhou 215021, China

Corresponding authors: Lichuan Wang (lcwang@suda.edu.cn) and Yan Chen (yanchen@suda.edu.cn)

This work was supported by the National Natural Science Foundation of China under Grant 61702352.

ABSTRACT The design of assembly line layout is one of the most important influential factors for the company's performance. The inappropriate arrangement of the order of workstations in the assembly line could cause the excessive movement of the material. In order to increase productivity and reduce production cost, a multi-objective optimization model with minimum the transfer distance of semi-finished products and the total area of assembly line was established in this paper. The Non-dominated Sorting Genetic Algorithm-II was applied to obtain the results of this model. The coding method, genetic operation and fitness function for three different kinds of sewing assembly line layouts (multi-line, U-shape two-line and U-shape three-line) were studied, while the workstations were organized in three ways: the order of processes, the type of machines, and the components of garment. The efficiency of the model was verified by the practice of sewing assembly lines for men's shirt. The layout schemes are compared with other algorithms. The results illustrated that NSGA-II algorithm is an effective tool to solve the sewing assembly line layout problem. The transfer distances of semi-finished products in the multi-line and U-shape (three-line) layouts arranged by garment components workstation layout were short. The multi-line layout arranged by the type of machines workstation layout occupied the smallest area. The model is suitable for solving the layout problem of the garment sewing workshop in practice, shortening the production cycle, and reducing the production cost. A variety of assembly line layout schemes are provided for apparel manufacturers in this paper. Apparel manufacturers can select the appropriate assembly line layout based on the actual conditions of the workshop and product.

INDEX TERMS Sewing assembly line layout, workstation layout, multi-objective optimization, non-dominated sorting genetic algorithm-II.

I. INTRODUCTION

With the rapid development of society, the demand of clothing products is more diversified. In order to quickly respond to the demands of multi-variety, small batch and short-cycle production, and effectively improve the production efficiency, the role of workshop logistics planning and layout in manufacturing system has become increasingly prominent [1]. The reasonable assembly line layout and logistic route improve the operation efficiency, production quality and reduce costs. The reasonable layout of assembly line can reduce the waste rate of production space, increase the utilization rate of the workshop area and expand the production scale. The reasonable planning of the logistics route can effectively shorten the transfer distance of semi-finished products, reduce the logistics cost, and improve the production efficiency of the workshop [2]. However, in practice, the assembly line layout is usually arranged based on the subjective experience of technicians. This empirical layout scheme has some disadvantages such as poor adaptability and low fluency of semi-finished products operation. It may result in low production efficiency and high production logistics cost, thus reducing the economic benefits of enterprises [3].

In the garment production process, most semi-finished products are in a state of stagnation or transfer, only 5% of the time is used for processing the semi-finished products. 20%-50% of operating expenses are used for handling and storage of semi-finished products [4]. In addition, such as

The associate editor coordinating the review of this manuscript and approving it for publication was Zhonglai Wang.

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see http://creativecommons.org/licenses/by/4.0/

oil stains, chromatic aberrations, etc., may be caused during the stagnation and transfer of semi-finished products [5]. The transfer of the semi-finished products requires time and operators. Reasonable arrangement of the semi-finished product transfer routes can shorten the transfer time, improve the production efficiency and the output [6].

The transfer of semi-finished products in garment sewing assembly line is complicated. There are great differences between the assembly lines based on the different organizational models. In order to improve the operation efficiency of semi-finished products in the sewing assembly line, it is very meaningful to study the layout of assembly line according to the actual situation of enterprises.

Generally, the assembly line layout can be divided into the five forms according to the arrangement shape and the flow pattern of semi-finished products.

(1) Straight layout is to arrange the workstations in a straight line, so that the semi-finished products transfer along the straight line from the first workstation during processing. But straight layout space is often narrow. It is suitable for assembly lines that the production process is short and simple, or the number of processes and machines is small.

(2) U-shape layout is to arrange the workstations according to the processing order, so that the logistics route is U-shape. This kind of layout is more compact and suitable for production workshops with small length and width.

(3) S-shape layout refers to the arrangement of workstations along the S-shape of logistics. It can be used to arrange longer process assembly line layouts in a larger shop area, such as automotive assembly line.

(4) The circular layout is to arrange the workstations along the circular shape. This layout is suitable to produce finished products that need to be returned to the starting point.

(5) Multi-line layout means that the workstations are arranged in multi lines. Semi-finished products can be transferred between different workstations perpendicular to the line or along the line. This layout can save the workshop space and the semi-finished products transfer paths are flexible. It is mainly applied to the assembly line layout with various kinds of processing products and complex processes.

There are many kinds of processes and machines involved in garment sewing assembly line. The multi-line layout, U-shape (two-line) layout and U-shape (three-line) layout of garment sewing assembly line are discussed in this research according to the characteristics of assembly line layout.

In general, production process of garment is relatively long, which also affects the transfer distance of semi-finished products. Fortunately, the workstations can be arranged reasonably by three different layouts based on production cycle. The order of processes workstation layout (PWL) is to arrange the workstations according to the processing order, as shown in Fig.1(a). The type of machines workstation layout (MWL) is that machines required for the same processing contents are arranged in same workstation, as shown in Fig.1(b). And the components of garment workstation layout (GWL) is to produce each garment component as a workstation, as shown in Fig.1(c).

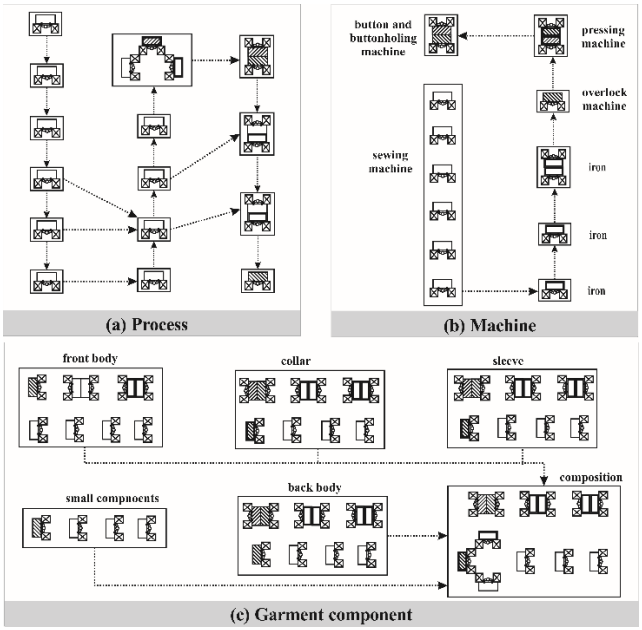


FIGURE 1. Modular diagram of three workstation layouts.

In this paper, the layout model of the apparel assembly line was analyzed by the practice of men's shirt manufacturing. The NSGA-II algorithm (Non-dominated Sorting Genetic Algorithm-II) algorithm was applied to obtain the layout schemes of different sewing assembly lines. The optimum way of the sewing assembly line in different workstation layouts could be found according to the results obtained from the model calculation. This paper was organized as follows: The assumptions, constraints and objective functions of the assembly line layout model were described in Section 2. The NSGA-II algorithm design was specifically described in Section 3. The men's shirt assembly line was used as an example, the different assembly line layouts for different workstations layouts and were optimized. Each optimization scheme was compared to each other to select an effective solution. A detailed experimental design and results were presented in Section 4. Discussion was illustrated in Section 5. In Section 6, several conclusions were summarized.

II. LITERATRUE REVIEW

The selection of appropriate optimization objectives and methods to solve layout problem has become a research hotspot. For example, Armal and Letchford [7] established a single-row facility layout optimization model to minimize material transfer distance. Ariafer and Ismail [8] studied U-shaped layout to minimize material handling costs between and within cells. Yang *et al.* [9] explored the specific solutions for logistics path problem in the layout of ring facilities and minimized the material handling cost. Kochhar [10] optimized the layout of multi-story workshops. According to characteristics of logistics path in multi-story workshops,

it was better to reduce the production costs from the horizontal material handling costs, and the vertical material handling costs.

It can be found that most researches focus on reducing the semi-finished product logistics distance and handling cost in the field of optimizing assembly line layout. There are few studies focus on the layout of garment sewing assembly lines that consider both semi-finished product transfer distance and assembly line area. In this research, a comprehensive garment sewing assembly line layout model is proposed, which considers the semi-finished product transfer distance and assembly line area.

The line layout problem is the determination of the most efficient physical arrangement of equipment in the factory floor of the manufacturing system to meet one or more objectives [11]. In recent years, several heuristic methods have been proposed in the literature to find the optimal solution of the production line, including particle swarm optimization [12], simulated annealing [13] and an ant system [14] and genetic algorithm (GA) [15]. The main disadvantage of these methods is that they do not explore enough possibilities when generating a solution, so random search algorithms are the only viable alternative, although they may only lead to near-optimal solutions.

Most of these research results are focus on single-objective optimization problems, while there are few studies on multi-target facility layout optimization. The solution to the multi-objective optimization problem is to transform the multi-objective optimization problem into a single-objective optimization problem, and then solve it by a single-objective genetic algorithm [16]–[18], which is not a true multi-objective optimization algorithm. Assembly line layout falls within the category of multi-objective optimization problems.

As with most issues, layout optimization requires consideration of multiple objectives that need to be optimized simultaneously with constraints. These problems are solved by multi-objective optimization techniques. In these problems, a set of several equally good (non-dominated) optimal solutions (rather than a single best advantage) is obtained and is called a Pareto set [19]. In fact, the optimal solution for Pareto concentration is becoming an increasingly effective method of determining the necessary trade-offs between conflict objective functions [20].

For many years, GA has been widely used for single objective functions [6] and multi-objective functions [21]. The advantage of genetic algorithm is that it does not require any initial guessing on the decision variables. GA uses several points of population as well as probability operators, namely reproduction, crossover, and variation obtained from natural genetics [22]. In addition, GA only uses the value of the objective function, instead by using any derivatives required by the gradient search technique [23]. Deb and colleagues developed a non-dominated sorting genetic algorithm (NSGA) to solve multi-objective optimization problems. NSGA-II introduces the concept of elitism [20]. An important feature of NSGA-II is that the best members are selected from the combined pool of parents and children (generated by crossing and mutation), and these become the parent of next generation [24].

Hosseini and Seifbarghy [25] proposed an integrated approach for dynamic facility layout problem considering the material handling equipment. The objectives of this problem were minimization of the fixed costs of MHE, minimization of material handling cost and minimization of machine rearrangement costs. The solution was obtained by NSGA-II algorithm. Chen *et al.* [26] used the NSGA-II algorithm to optimize the design of the steam generator and the primary loop of nuclear power plant. Bhattacharya and Bandyopadhyay [27] presented an energy recovery facility planning method considering energy loss, transport performance and space demand. NSGA-II algorithm was used to obtain the optimal layout. Yang *et al.* [28] combined the energy mixing facility model with the NSGA-II algorithm as a multi-objective optimal facility allocation model to allocate renewable energy facilities on the roof of campus buildings.

The main purpose of this research work is to optimize the layout of garment sewing assembly lines by using NSGA-II. The objectives of the optimization include reducing material transfer distance and production line area. The validity of the algorithm is verified by the application of the existing actual running factory data.

III. MODEL FORMULATION

The quality of assembly line layout of the workshop affects the transfer distance of semi-finished products and the total area of workshop [29]–[31]. The amount of material transferred depends on the matching degree between product processes and workstations [32], [33]. The total area of assembly line depends on the positional relationship between the workstations [34]–[36]. The enterprises hope that the semi-finished products are transferred in the shortest distance, while workstations require the minimum total area. Therefore, in this paper, a multi-objective optimization model considering the above two factors is established.

A. ASSUMPTIONS

In this paper, the mathematical models are supported by literature reference [37]–[39], to maintain the reasonable and scientific nature of the hypothesis. The assumptions for the establishment of multi-objective model are listed as follows:

1) The workstations are rectangles, and their lengths and widths are known. The workstation can be arranged in two ways: its length direction is parallel to the length of workshop; its width direction is parallel to the length of workshop.

2) The centers of workstations on the same row are in the same horizontal line.

3) There is a minimum separation requirement between different workstations.

4) All workstations are sequentially arranged in the order of the length of workshop.

B. PARAMETER VARIABLES

According to assumptions and requirements of the modeling, constraints are proposed. Calculation formulas of four vertexes coordinates of the smallest rectangle including all workstations are shown in Eq (1), (2), (3), (4).

$$x_a = min\left\{x_i - \frac{1}{2}\left[C_i \times L_i + (1 - C_i)\, W_i\right]\right\} \tag{1}$$

$$x_b = max\left\{x_i + \frac{1}{2}\left(C_i \times L_i + (1 - C_i)\, W_i\right)\right\} \tag{2}$$

$$y_a = min\left\{y_i - \frac{1}{2}\left(C_i \times W_i + (1 - C_i)\, L_i\right)\right\} \tag{3}$$

$$y_b = max\left\{y_i + \frac{1}{2}\left(C_i \times W_i + (1 - C_i)\, L_i\right)\right\} \tag{4}$$

where, i is the index of workstation, $i = 1, 2 \ldots, N$; N is the total number of workstations; x_i is the center point abscissa of the workstation i; y_i is the center point ordinate of the workstation i; C_i is the placement direction of workstation i; L_i is the length of the workstation i; W_i is the width of the workstation i; x_a is the abscissa of the lower-left vertex of the smallest rectangle including all workstations; x_b is the abscissa of the upper-right vertex of the smallest rectangle including all workstations; y_a is the ordinate of the lower-left vertex of the smallest rectangle including all workstations; y_b is the ordinate of the upper-right vertex of the smallest rectangle including all workstations.

When workstation i is the first workstation in the first row, the workstation center coordinates are calculated as follows:

$$x_i = d_0 + \frac{1}{2}\left(C_i \times L_i + (1 - C_i)\, W_i\right) \tag{5}$$

$$y_i = h_0 + \frac{1}{2}\left(C_i \times W_i + (1 - C_i)\, L_i\right) \tag{6}$$

where, d_0 is the minimum horizontal distance between the workstation and workshop boundary (width of workshop), unit: m; h_0 is the minimum vertical distance between the workstation and workshop boundary (length of workshop), unit: m.

Constrains of binary conditions of variables are shown in Eq (7) and Eq (8).

$$C_i \in (0, 1) \tag{7}$$

where, $C_i = 1$ means the long side of the workstation i is parallel to the length of workshop; $C_i = 0$ means the long side of the workstation i is parallel to the width of workshop.

$$Z_{ir} \in (0, 1) \tag{8}$$

where, $Z_{ir} = 1$ indicates that the workstation i is on the r_{th} line; $Z_{ir} = 0$ indicates that the workstation i is not on the r_{th} line.

C. CONSTRAINTS

In this research, the layouts of the assembly lines are divided into three types: multi-line, U-shape (two-line), U-shape (three-line), which are shown in Fig. 2 to Fig. 4.

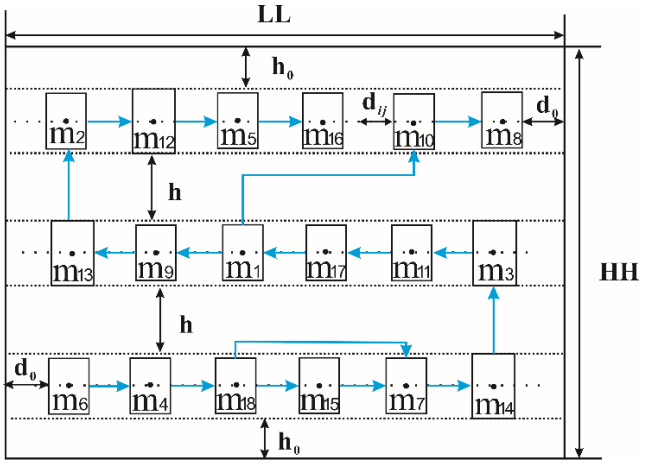


FIGURE 2. Sewing assembly line diagram of multi-line layout.

As shown in Fig. 2, the assembly line has more than two rows. Eq (9) ensures that each workstation is placed at one location.

$$\sum_{r=1}^{R} Z_{ir} = 1 \tag{9}$$

where, r is the index of rows, $r = 1, 2 \cdots, R$; R is the total number of rows; Z_{ir} is the workstation i in the r_{th} row.

Eq (10) guarantees that the number of workstations in a row does not exceed the total number of workstations.

$$\sum_{i=1}^{N} Z_{ir} \leq N \tag{10}$$

Eq (11) ensures that there is no overlap or intersection between workstations in the same row.

$$|x_{i+1} - x_i| Z_{(i+1)r} Z_{ir} \geq \frac{\left[C_{i+1} \times L_{i+1} + (1 - C_{i+1})\, W_{i+1}\right] + \left[C_i \times L_i + (1 - C_i)\, W_i\right]}{2} + d_{i(i+1)} \tag{11}$$

where, d_{ij} is the minimum lateral spacing that must be maintained between two adjacent workstations i and j, unit: m.

Eq (12) guarantees that the center points of workstations in the same row are on the same horizontal line.

$$y_{r(i+1)} Z_{(i+1)r} = y_{ri} Z_{ir} \tag{12}$$

Eq (13) ensures that there is no overlap or intersection between workstations of two adjacent rows.

$$|y_j - y_e| Z_{j(r+1)} Z_{er} \geq \frac{\left[C_i \times W_i + (1 - C_i)\, L_i\right] + \left[C_j \times W_j + \left(1 - C_j\right) L_j\right]}{2} + h \tag{13}$$

where, j and e are the indexes of workstation, j, $e = 1, 2 \ldots, N$; y_j is the center point ordinate of the workstation j; y_e is the center point ordinate of the workstation e; h is the spacing between two rows, unit: m.

Eq (14) ensures that the sum of the total length of the workstations in a row and the minimum horizontal spacings of the workshop boundaries is less than the length of the workshop.

$$|x_1 - x_n| Z_{1r} Z_{nr} + 2d_0 + \frac{[C_1 \times L_1 + (1-C_1)\, W_1] + [C_n \times L_n + (1-C_n)\, W_n]}{2} \leq \mathrm{LL} \tag{14}$$

where, LL is the length of the workshop, unit: m; n is the index of the last workstation in a row, $n = 1, 2 \ldots, N$.

Eq (15) guarantees that the length of the minimum rectangle including all workstations does not exceed the length of the workshop.

$$|x_b - x_a| + 2d_0 \leq \mathrm{LL} \tag{15}$$

Eq (16) ensures that the width of the minimum rectangle including all workstations does not exceed the width of the workshop.

$$|y_b - y_a| + 2h_0 \leq \mathrm{HH} \tag{16}$$

where, HH is the width of the workshop, unit: m.

As shown in Fig. 3, when the total length of workstations in a row exceeds the length of workshop, the workstations automatically turn around to form a U-shaped (two-line) mode. The constraints are the same as multi-line.

As shown in Fig. 4, $K+1$ is assumed to be the number of workstations in the first row ($1 < K < N-2$). When the $(K+2)_{th}$ workstation is placed in the first row and the total length of the workstations exceeds the length of workshop, the $(K+2)_{th}$ workstation is placed on the second row. And the remaining workstations are placed on the third row. Eq (17) calculates the abscissa of workstation center point.

$$x_i = \begin{cases} x_{i-1} + \dfrac{[C_{i-1} \times L_{i-1} + (1-C_{i-1})\, W_{i-1}]}{2} + \dfrac{[C_i \times L_i + (1-C_i)\, W_i]}{2} + d_{i(i-1)}, & 1 < i \leq K+1 \\ x_{i-1} - \dfrac{[C_i \times L_i + (1-C_i)\, W_i]}{2} - \dfrac{[C_i \times L_i + (1-C_i)\, W_i]}{2} - d_{i(i-1)}, & K+1 < i \leq N \end{cases} \tag{17}$$

Eq (18) calculates the ordinate of workstation center point. The remaining constraints are the same as multi-line.

$$y_i = \begin{cases} \dfrac{1}{2}[C_i \times W_i + (1-C_i)\, L_i] + h_0, & 1 < i \leq Q \\ y_{i-1} + \dfrac{[C_{i-1} \times W_{i-1} + (1-C_{i-1})\, L_{i-1}]}{2} + \dfrac{[C_i \times W_i + (1-C_i)\, L_i]}{2} + h, & K+1 \leq i \leq N \end{cases} \tag{18}$$

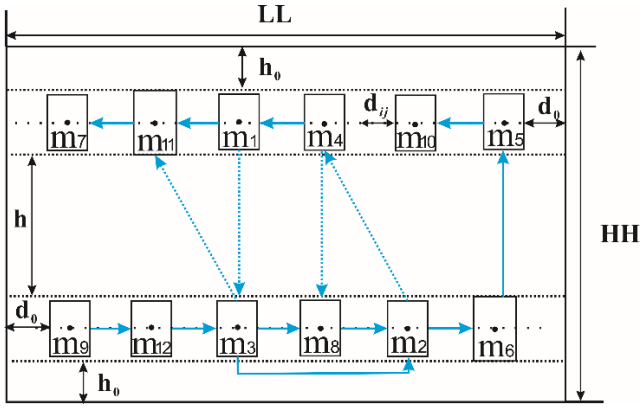


FIGURE 3. Sewing assembly line diagram of U-shape (two-line) layout.

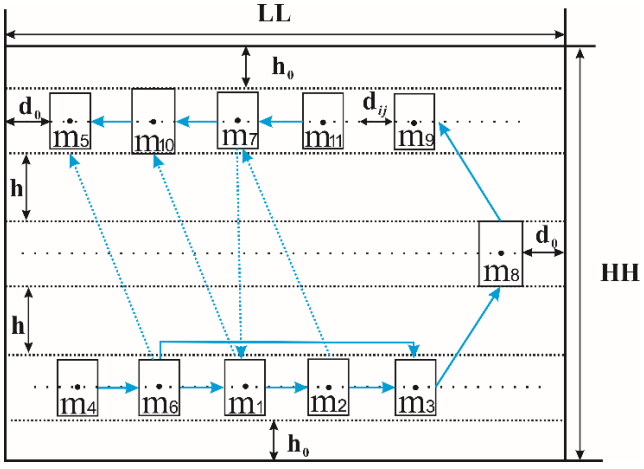


FIGURE 4. Sewing assembly line diagram of U-shape (three-line) layout.

D. OBJECTIVE FUNTICTION

Objective function 1: Minimizing the transfer distance of the semi-finished products.

$$\mathrm{F} = \sum_{i=1}^{n} \sum_{j=1}^{n} Q_{ij} D_{ij} \tag{19}$$

where, F is the total transfer distance of semi-finished products, unit: km; Q_{ij} is the transfer frequency of semi-finished products from workstation i and workstation j; D_{ij} is the broken line distance or linear distance between workstation i and workstation j, unit: m.

Objective function 2: Minimizing the area occupied by the sewing assembly line.

$$\mathrm{S} = (x_b - x_a) \times (y_b - y_a) \tag{20}$$

where, S is the area of the assembly line including all workstations, unit: m^2.

IV. MATHODLOGY

The NSGA-II algorithm is applied to efficiently obtain the optimization results of multi-objective assembly line layout. The NSGA-II algorithm can be simply divided into seven

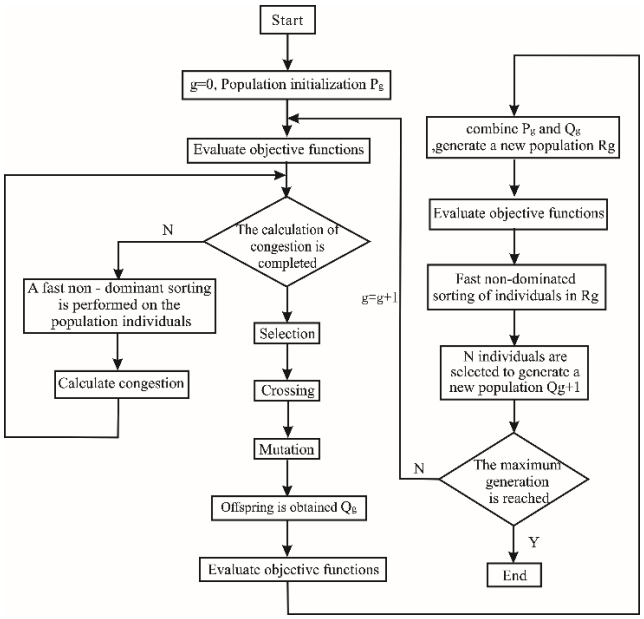


FIGURE 5. The flow chart of NSGA-II algorithm.

steps: (1) coding, (2) population initialization, (3) calculation of fitness value, (4) non-dominated sorting and calculation of congestion, (5) selecting operation, (6) crossing operation (7) mutation operation. The flow chart of NSGA-II algorithm is shown in Fig. 5. The specific processes are listed as follows.

Step 1: The workstations are coded with real numbers.

Step 2: The overall size is set, and the initial population is generated according to the constraints.

Step 3: According to the objective function value of each individual, the rapid non-dominant sorting of the contemporary population and the congestion distance are calculated.

Step 4: Individuals are selected by the tournament mechanism based on the results of the ranking and congestion distance calculations. And then individuals are crossed and mutated.

Step 5: The elite retention strategy is implemented. The parent and child are combined. A new generation of populations is generated by selecting a combined population by fast non-dominant sorting and virtual congestion distance.

Step 6: The number of iterations is increased by 1, and step 3 is returned. This loop will continue until the maximum number of iterations is reached.

A. CODING

The order of the workstations is coded by using natural number. Each workstation corresponds to a gene position in the chromosome. The chromosome coding method is listed as follows.

$$\{m_1, \cdots, m_i, \cdots, m_n\}$$

According to Fig. 2, Fig. 3 and Fig. 4, the chromosomes of three different assembly line layouts (multi-line, U-shape two-line, U-shape three-line) are shown in Fig. 6.

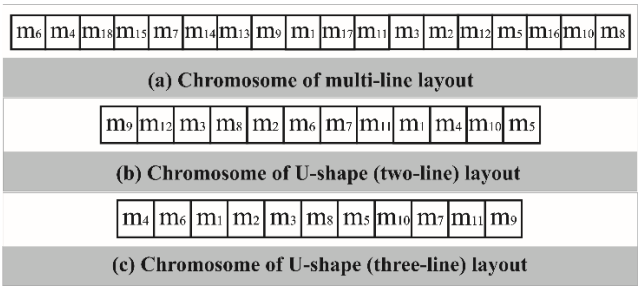


FIGURE 6. Schematic diagrams of coding for different assembly line layouts: (a) Chromosome of Multi-line layout; (b) Chromosome of U-shape (two-line) layout; (c) Chromosome of U-shape (three-line) layout.

B. POPULATION INITIALIZATION

In this research, the population initialization process is implemented in three steps:

Firstly, the row number of workstations is randomly generated, and the workstation numbers of each row are obtained. The sorting of generated workstation numbers is broken. The minimum position on the bottom left corner and the maximum position on the top right corner of the smallest rectangle including all workstations are generated. Based on the maximum position and the minimum position, the abscissas of the workstations in these two rows are generated.

Then, the workstations are sorted from left to right based on the abscissas. From the leftmost position, the abscissa of the first workstation is compared with that of the second workstation, and then the abscissa of the second workstation is compared with that of the third workstation. The abscissas of two adjacent workstations in the row are compared to determine whether there is a crossover. It is judged whether the center distance between two adjacent workstations is less than the sum of 0.5 times the lengths of the two workstations and the security interval d_0. If the center distance between two adjacent workstations is greater than the sum of 0.5 times the lengths of the two workstations and the security interval, there is no crossover.

As shown in Fig. 7, d is assumed to be the center distance of two adjacent workstations (m_i, m_j), $C_i = 1$, $C_j = 0$.

$$d \leq 0.5 \times \left(L_i + W_j\right) + d_{ij} \tag{21}$$

As shown in Fig. 7, m_i is the leftmost workstation in the workshop, m_e is the rightmost workstation in the workshop. $C_i = 1$, $C_e = 1$. The constraint is that x_i needs to satisfy is as follows:

$$x_i \geq 0.5 \times L_i + d_0 \tag{22}$$

Another constraint is that x_e needs to satisfy is as follows:

$$x_e \leq \mathrm{LL} - 0.5 \times L_e - d_0 \tag{23}$$

Finally, *temp0*(*temp0*=0) is generated as a ruler. The height of first row at the bottom layer is the sum of *temp0* and h_0, which give the height of all the workstations in each row. The ordinate of workstations in fist row is the sum of the height of first row and 0.5 times of the maximum height of

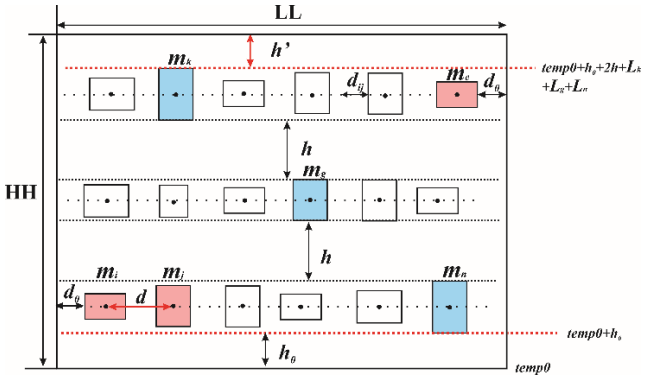


FIGURE 7. Diagram of assembly line layout.

workstations in first row. If it is not the last row, the new ruler should be the sum of *temp0* and the maximum height of the workstation in the row and the row spacing h. If it is the last row, the new ruler should be the sum of *temp0* and the maximum height of the workstation in the row. The difference between the width of the workshop and the rule is determined to satisfy the constraint. If it is not less than h_0, the constraints are satisfied.

As shown in Fig. 7, m_n, m_g, m_k is the highest workstation in each row, $h^{'}$ is the distance between the edge of the highest station and the shop floor in the last row. $C_n = 0$, $C_g = 0$, $C_k = 0$. The constraint that $h^{'}$ needs to satisfy is as follows.

$$h^{'} = LL - temp0 - h_0 - 2h - L_n - L_g - L_k \leq h_0 \quad (24)$$

C. CALCULATION OF FITNESS VALUE

The abscissa, ordinate and number of each workstation in each row are obtained by decoding. Then the Hamming distance and the Euclidean distance are calculated. It is looped in turn, resulting in x_a, x_b, y_a, y_b.

D. NON-DOMINATED SORTING AND CALCULATION OF CONGESTION

The fast non-dominated sorting solution is found and classified into a first-level non-dominated solution. After that, a new non-dominated solution is found in the remaining solutions and classified into a second-level non-dominated solution. The loop will be running in turn until all solutions are assigned.

When the ranks of the non-dominated solutions are the same to each other, it is necessary to distinguish the advantages and disadvantages, according to the crowding distance of the non-dominated solutions. And the evenly distributed solution is preferentially selected since the relatively large crowding distance.

E. SELECTING OPERATION

The selected population evolution is close to the position of the non-dominated optimal solution for the purpose of the average dispersion of the non-dominated solution frontier. The method of binary tournament selection is to arbitrarily select two non-dominated solutions (i and j) and put them into the mating pool. The individual i is better than j when the non-dominated level $rank_i < rank_j$; if $rank_i = rank_j$, the congestion distance was $d\,(i)_{distance} > d\,(j)_{distance}$. Elite retention strategies are adopted to accelerate the rate of convergence.

F. CROSSING OPERATION

Deb and Agrawal found that the progeny individuals produced by single-point crossover had the same geometric center as their parents. Thus, they designed the simulated binary crossover (SBX) to make real-coded crossover operator has this property. Therefore, simulated binary crossover (SBX) is used for multi-objective optimization of assembly line layout, as follows:

$$\begin{aligned} x^{k+1}(1) &= 0.5\left((1-\alpha)\,x^k(1) + (1+\alpha)\,x^k(2)\right) \\ x^{k+1}(2) &= 0.5\left((1+\alpha)\,x^k(1) + (1-\alpha)\,x^k(2)\right) \end{aligned} \quad (25)$$

$$\alpha = \begin{cases} (2\beta)^{\frac{1}{\mu+1}}, & \beta < 0.5 \\ (2\,(1-\beta))^{\frac{1}{\mu+1}}, & \beta \geq 0.5 \end{cases} \quad (26)$$

where, x^k (1) and x^k (2) are the pre-crossover parent individual genes; x^{k+1} (1) and x^{k+1} (2) are individual genes of the progeny after crossing; β is a uniformly distributed random number, $\beta\epsilon$ (0, 1); μ is a cross-distribution index, generally $\mu = 20$.

G. MUTATION OPERATION

Polynomial mutation strategy is applied to perform the mutation operation. If the gene position v_i (the i_{th} center) of a chromosome is selected for mutation, a random number δ between (0, 1) is first generated, and then a new cluster center is generated by the following formula:

$$v_i^{'} = v_i + \begin{cases} (2\delta)^{\frac{1}{\eta_m+1}} - 1, & \delta < 0.5 \\ 1 - [2\,(1-\delta)]^{\frac{1}{\eta_m+1}}, & \delta \geq 0.5 \end{cases} \quad (27)$$

where, η_m is the variation distribution index, generally $\eta_m = 20$. In this research, polynomial variation is performed with 50% probability, and a new individual is regenerated with 50% probability.

V. CASE STUDY

The sewing assembly lines under different layout schemes were analyzed by the model established. The transfer distance of semi-finished products, and the area of sewing assembly line of men's shirt were studied on the results of analysis.

A. EXPERTMENT DATA

As shown in Fig. 8, there are 48 processes in the production of the men's shirt. The workstations are arranged in

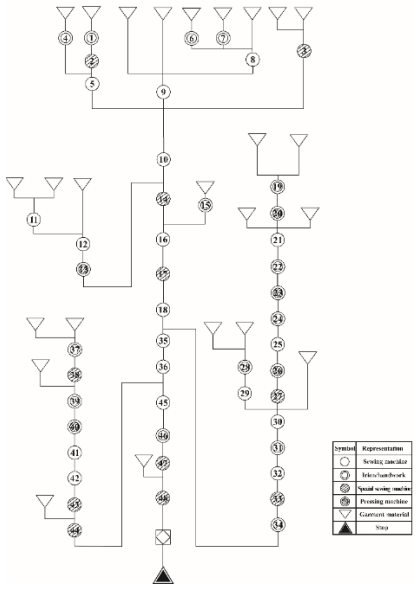


FIGURE 8. Process flow chart of the production of men's shirt.

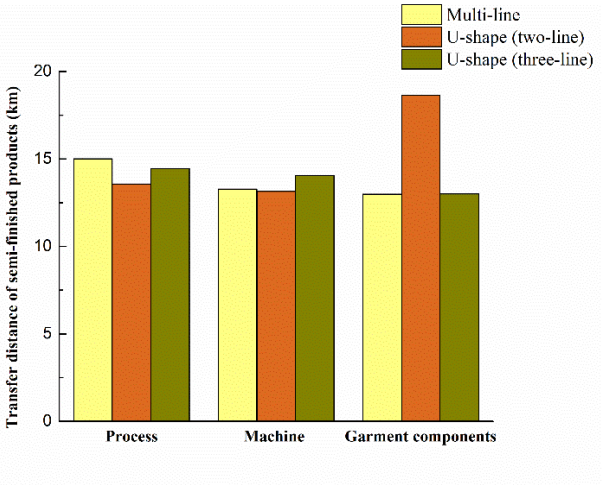


FIGURE 9. Transfer distances of semi-finished products in different assembly lines.

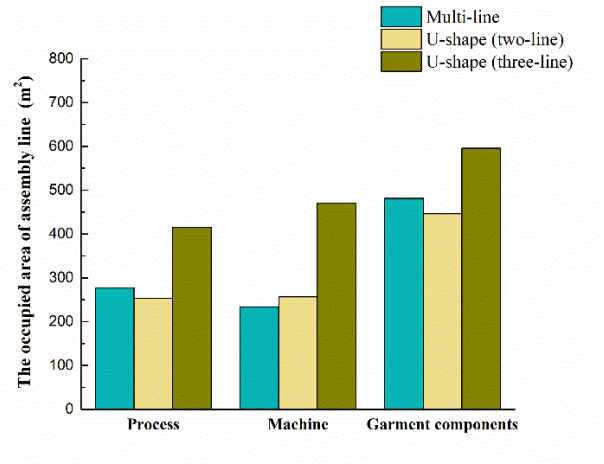


FIGURE 10. Areas of different assembly lines.

three different layouts according to the production cycle, as shown in Table 1. The type of sewing machine included in the workstation, the number of sewing machines in the workstation, directions, lengths and widths of workstations under different layouts were shown in Table 1.

As shown in Table 1, the men's shirt sewing assembly line can be divided into 14 workstations and 26 sewing machines according to workstation layouts of process and machine. According to the workstation layout of garment components, different machines need to be arranged in the workstation, so there are more sewing machines required. And there are 15 workstations and 41 sewing machines in the sewing assembly line.

B. PARAMETER SETTING

The length of the workshop was 100m and the width was 50m. The minimum distance must be maintained between two adjacent rows of workstations was 2.5m. The minimum lateral spacing between the workstation and the workshop boundary was 4m. The minimum longitudinal spacing between the workstation and the workshop boundary was 2m. The minimum horizontal distance between the same equipment was 0.7m. The minimum lateral distance between different equipment was 1m. The initial population size of the NSGA-II algorithm was 200, and its genetic algebra was 500.

C. ANALYSIS OF RESULTS

After calculation, the layout optimization schemes for the assembly lines of men's shirt were shown in Table 2.

As shown in Fig. 9 and Fig. 10, for the assembly lines of PWL, the semi-finished products of the U-shape (two-line) layout had the shortest total transfer distance and occupied the smallest area, the semi-finished products of multi-line layout had the longest total transfer distance and occupied the largest area. For the assembly lines of MWL, the semi-finished products of U-shape (two-line) layout had the shortest total transfer distance, the total transfer distance of semi-finished products of multi-line layout was close to that of the U-shape (two-line) layout, but the area of the multi-line layout was the smallest. For the assembly lines of GWL, the total transfer distance of the semi-finished products of the multi-line and U-shape (three-line) layout were shorter than that of U-shape (two-line), but the area of the U-shape (two-line) was smallest. Among the nine optimized assembly lines, the total transfer distances of the semi-finished products of the multi-line and U-shape (three-line) layout arranged by GWL were the shortest (13km), while the area of sewing assembly line of multi-line layout arranged by MWL was the smallest (234m^2). In the contrast, the semi-finished products of the U-shape (two-line) arranged by GWL had the longest total transfer distance(18.6km), while it occupied the largest area (596m^2).

TABLE 1. Workstation data for different workstation layouts of men's shirt assembly line.

Process				
Workstation number	Equipment	Number of equipment	$L_i \times W_i$ /m×m	C_i
1	Electric steam iron	1	1.6×1.4	0
2	Electric steam iron, Pocket creasing machine, Trademark shearing machine, Manual operation	4	5.8×3	1
3	Electric steam iron, Manual operation	2	2.8×1.6	1
4	Buttonhole stitching machine, Upper and lower cutting machine, Shearing collar machine	3	4.1×2.8	1
5	Button attaching machine, Collar turning machine	2	2.3×1.6	1
6	Five-thread overlock machine	1	1.6×1.2	0
7	Sewing machine	1	1.6×1.2	0
8	Sewing machine	1	1.6×1.2	0
9	Sewing machine	1	1.6×1.2	0
10	Utility press, Fusing press machine, Collar former, Shirt collar former, Styling press machine	5	9×4.4	1
11	High speed Sewing machine, Sewing machine	2	2.4×1.6	1
12	Sewing machine	1	1.6×1.2	0
13	Sewing machine	1	1.6×1.2	0
14	Sewing machine	1	1.6×1.2	0
Machine				
Workstation number	Equipment	Number of equipment	$L_i \times W_i$ /m×m	C_i
1	Sewing machine	1	1.6×1.2	0
2	Sewing machine	1	1.6×1.2	0
3	Sewing machine	1	1.6×1.2	0
4	Sewing machine	1	1.6×1.2	0
5	High speed Sewing machine, Sewing machine	2	2.4×1.6	1
6	Sewing machine	1	1.6×1.2	0
7	Sewing machine	1	1.6×1.2	0
8	Electric steam iron	1	1.6×1.4	0
9	Electric steam iron, Trademark shearing machine, Manual operation	3	4.4×2.9	1
10	Electric steam iron, Pocket creasing machine, Manual operation	3	4.4×2.9	1
11	Buttonhole stitching machine, Collar turning machine	2	2.3×1.6	1
12	Five-thread overlock machine	1	1.6×1.2	0
13	Button attaching machine, Upper and lower cutting machine, Shearing collar machine	3	4.1×2.8	1
14	Utility press, Fusing press machine, Collar former, Shirt collar former, Styling press machine	5	9×4.4	1
Garment component				
Workstation number	Equipment	Number of equipment	$L_i \times W_i$ /m×m	C_i
1	Electric steam iron, Pocket creasing machine	2	2.7×1.6	1
2	Sewing machine	1	1.6×1.2	0
3	Buttonhole stitching machine, Button attaching machine, Trademark shearing machine, Sewing machine	4	5.7×2.8	1
4	Electric steam iron, Sewing machine	2	2.6×1.6	1
5	Fusing press machine, Sewing machine, Shirt collar former, Electric steam iron,	4	6.5×4.4	1
6	Collar turning machine, Collar former, Upper and lower cutting machine, Sewing machine	4	6.4×2.8	1
7	High speed Sewing machine, Electric steam iron, Shearing collar machine, Manual operation	4	5.7×3	1
8	Utility press, Sewing machine, Electric steam iron	3	4×3	1
9	Sewing machine, Electric steam iron, Button attaching machine	3	4×3	1
10	Sewing machine, Fusing press machine, Buttonhole stitching machine	3	4×3.7	1
11	Sewing machine, Manual operation	2	2.6×1.6	1
12	Sewing machine, Button attaching machine	2	2.4×1.6	1
13	Five-thread overlock machine, Trademark shearing machine, Sewing machine, Manual operation	3	4×2.9	1
14	Sewing machine, Five-thread overlock machine	2	2.4×1.6	1
15	Sewing machine, Buttonhole stitching machine	2	2.4×1.6	1

TABLE 2. Optimization results of different assembly line layouts.

Assembly line layout	Workstation layout	Row number	Sorting of workstations
Multi-line	Process	1	{11, 3}
		2	{8, 4, 5, 1}
		3	{13, 14, 12, 7, 6, 9}
		4	{2, 10}
	Machine	1	{5, 10, 9}
		2	{2, 3, 7, 8, 4}
		3	{6, 12, 11, 1}
		4	{13, 14}
	Garment components	1	{5, 4,, 7,, 1}
		2	{10, 6, 15, 2, 12}
		3	{14, 9, 3}
		4	{8, 13, 11}
U-shape (two-line)	Process	1	{3, 13, 1, 11, 5, 12, 14, 8, 6}
		2	{7, 2, 4, 10, 9}
	Machine	1	{1, 12, 4, 8, 7, 5, 11, 3, 6, 2}
		2	{14, 9, 13, 10}
	Garment components	1	{9, 10, 11, 4, 2, 12, 15, 14}
		2	{8, 6, 7, 5, 3, 13, 1}
U-shape (three-line)	Process	1	{14, 12, 5, 1, 11, 13, 3, 9, 6}
		2	{8}
		3	{7, 4, 2, 10}
	Machine	1	{3, 10, 6, 9, 12, 1, 4}
		2	{8}
		3	{5, 2, 11, 7, 13, 14}
	Garment components	1	{13, 14, 7, 2, 3, 1, 11}
		2	{8}
		3	{5, 4, 6, 15, 12, 9, 10}

VI. COMPARISONS WITH OTHER APPROACHES

Many researchers have proved the high computational complexity of different layout design problems. Many heuristic algorithms [40], [41] and meta heuristic algorithms have been proposed to solve single-row and multi-line layout design problems. Nevertheless, genetic algorithm (GA) is the most popular used method to solve the problem of assembly line layout [21], [42], [43]. In most cases, it has been shown that GA concludes better solutions than other metaheuristics methods in assembly line layout problem [44]–[46]. The solution found by the NSGA-II algorithm was compared with the optimal solution of the adaptive genetic algorithm (AGA) and the genetic simulated annealing algorithm (GA-SA).

Combined optimization objective function of assembly line layout.

$$C_{min}=\alpha_1\beta_1\sum_{i=1}^{n}\sum_{j=1}^{n}Q_{ij}D_{ij}K+\alpha_2\beta_2\,(x_b-x_a)\times(y_b-y_a) \tag{28}$$

where, α_1 and α_2 are normalization factors. In order to ensure the dimensions of the two parts of the function are unified, the calculation formulas of α_1 and α_2 are as follows.

$$\alpha_1=\frac{1}{\sum_{i=1}^{n}\sum_{j=1}^{n}Q_{ij}D_{ij}K}$$
$$\alpha_2=\frac{1}{(x_b-x_a)\times(y_b-y_a)} \tag{29}$$

where, β_1 and β_2 are weighting factors, they are determined by the degree of emphasis on the semi-finished products transfer and assembly line area in actual production. The weighting condition is satisfied $\beta_1+\beta_2=1$. In this research, $\beta_1=\beta_2=0.5$.

The comparison results between the different approaches are shown in Table 3. The best values achieved for each problem are highlighted in bold.

As shown in Table 3, the results indicate that the NSGA-II algorithm obtains more effective solutions than other algorithms. In the solutions of the AGA algorithm, three types of assembly lines have the smallest production line area, but the transfer distance of the semi-finished products is longer than that obtained by the other two algorithms. In the solutions of the GA-SA algorithm, the transfer distance of the

TABLE 3. Comparison results between the NSGA-II algorithm and other algorithms.

Assembly line layout	Workstation layout	NSGA-II		GA-SA		AGA	
		Transfer distance/km	Area /m^2	Transfer distance/km	Area /m^2	Transfer distance/km	Area /m^2
Multi-line	Process	**15**	**277**	16.7	302	20	291
	Machine	**13.3**	**234**	14.9	269	17.5	274
	Garment components	**13**	481	14.3	456	17.1	**434**
U-shape (two-line)	Process	**13.6**	**253**	15.4	281	17.9	289
	Machine	**13.1**	**257**	15	276	17.1	284
	Garment components	18.6	446	**16.8**	452	22	**431**
U-shape (three-line)	Process	**14.4**	**415**	15.8	434	16.4	426
	Machine	**14**	**471**	15	442	16.2	463
	Garment components	**13**	596	14.8	613	15.4	**517**

semi-finished products of the U-shaped (two-line) under the arrangement of garment components is the shortest, but the assembly line area is larger than that obtained by the other two algorithms. In the solutions of the NSGA-II algorithm, eight types of assembly lines have short transfer distance of the semi-finished products and small area. Although the AGA algorithm and the GA-SA algorithm improve the GA algorithm, this method converts the multi-objective optimization problem into a single-objective optimization problem, and indirectly solves it by using the mature single-objective optimization method. The weight vectors of the objectives are determined in advance, and each objective has a different dimension. The weight coefficients are used to linearly combine multiple objectives with different physical meanings. And the weight coefficients of the objective function are subjective and unscientific. In fact, the optimal solution obtained by this method is only one solution in the optimal solution set, so it is difficult to obtain the optimal optimization effect. Therefore, the assembly line layouts generated by NSGA-II algorithm are significantly better than that obtained by other algorithms. The comparison results also illustrate that the transfer distances of semi-finished products in the multi-line and U-shape (three-line) layouts arranged by GWL were short. The multi-line layout arranged by MWL occupied the smallest area.

VII. DISCUSSION

As shown in the Fig. 2, Fig. 3 and Fig. 4, it can be concluded that multi-line layout and U-shaped (two-line) layout can arrange more workstations in a certain space, while the U-shaped (three-line) layout requires more space to arrange the same number of workstations. Most importantly, the multi-line and U-shaped (two-line) layout will be better types of sewing assembly line for companies with restricted areas, while companies want to arrange more machines in their factories.

As shown in Table 1, in PWL and MWL, there are seven workstations need one machine. In GWL, there is only one workstation need one machine and six workstations with two machines. Therefore, the area of three assembly lines of GWL is larger than that of PWL and MWL.

Generally, the workstations could be arranged by three different ways (PWL, MWL and GWL) based on different layouts of assembly lines. Different from the random arrangement of the workstations, the three kinds of workstation layouts (PWL, MWL and GWL) were more reasonable, they are not only saved semi-finished products transfer distances but also reduced production costs [47]. According to PWL, the semi-finished product has short transfer distance and production cycle. It is suitable for producing a single variety of garment products. According to MWL, the transfer distance of semi-finished products is shorter than that of PWL, but the machines can serve different kinds of garment, while the utilization rate of sewing machines is high, so it is suitable for producing garment with varied styles. According to GWL, the transfer distance of the semi-finished product is short, and the production cycle is short, while the reaction speed is fast. It is suitable for producing multi-variety and small batch of garment [48]–[50].

VIII. CONCLUSION

In this research, NSGA-II algorithm is proposed to solve sewing assembly line layout problem. The approach considers different types of assembly lines (multi-line, U-shape two-line, and U-shape three-line) of manufacturing layout environment, while the workstations are organized in three ways (the order of processes, the type of machines, and the components of garment). The optimal layout of assembly lines is obtained by NSGA-II algorithm, which minimizes the transfer distance of semi-finished products and the area of assembly line. The results are also compared with the optimal assembly line layouts obtained by GA-SA algorithm

and AGA algorithm. The comparison results indicate that the NSGA-II algorithm is more effective and has a higher chance of obtaining the best solution for the sewing assembly line layout problem. The multi-line and U-shape (three-line) assembly lines arranged by GWL would be better ways to arrange the workstations in the factory to save area and reduce semi-finished products transfer distances. PWL is suitable for producing a single variety of garment products. MWL is suitable for producing garment with varied styles. GWL is suitable for producing multi-variety and small batch of garment. Apparel manufacturers can select the corresponding assembly line layout based on the conditions of product and workshop. The research design and research methods of the model can also be applied to other industries.

There are many studies on the optimization of assembly line layout in the automotive industry and other fields, which is a blank in the garment industry. It should be the first time that NSGA-II algorithm has been applied to the layout optimization of garment sewing assembly line. Other algorithms are used to optimize the layout of garment sewing assembly line, which is the content of our future research. Production systems that respond quickly to market changes and cost-effective responses are becoming more and more popular in global enterprises. Alternative layouts are generated by using simulation techniques to predict he fluency and efficiency of the production line, which is undoubtedly another promising area of research.

REFERENCES

[1] H. Long, Z. Zhang, Z. Song, and A. Kusiak, "Formulation and analysis of grid and coordinate models for planning wind farm layouts," *IEEE Access*, vol. 5, pp. 1810–1819, 2017. doi: 10.1109/ACCESS.2017.2657638.

[2] A. Tayal and S. P. Singh, "Integrating big data analytic and hybrid firefly-chaotic simulated annealing approach for facility layout problem," *Ann. Oper. Res.*, vol. 270, nos. 1–2, pp. 489–514, Nov. 2008.

[3] A. Diponegoro and B. R. Sarker, "Flow distance reduction for a multi-product flowline with sets of identical machines," *Eur. J. Oper. Res.*, vol. 147, no. 3, pp. 591–612, Jun. 2003.

[4] W. Yi, H.-L. Chi, and S. Wang, "Mathematical programming models for construction site layout problems," *Automat. Construct.*, vol. 85, pp. 241–248, Jan. 2018.

[5] C. Friedrich, A. Klausnitzer, and R. Lasch, "Integrated slicing tree approach for solving the facility layout problem with input and output locations based on contour distance," *Eur. J. Oper. Res.*, vol. 270, no. 3, pp. 837–851, Nov. 2018.

[6] A. M. El-Baz, "A genetic algorithm for facility layout problems of different manufacturing environments," *Comput. Ind. Eng.*, vol. 47, nos. 2–3, pp. 233–246, Nov. 2004.

[7] A. R. S. Amaral and A. N. Letchford, "A polyhedral approach to the single row facility layout problem," *Math. Program.*, vol. 141, nos. 1–2, pp. 453–477, Oct. 2013.

[8] S. Ariafar and N. Ismail, "An improved algorithm for layout design in cellular manufacturing systems," *J. Manuf. Syst.*, vol. 28, no. 4, pp. 132–139, Dec. 2009.

[9] T. Yang, B. A. Peter, and M. Tu, "Layout design for flexible manufacturing systems considering single-loop directional flow patterns," *Eur. J. Oper. Res.*, vol. 164, no. 2, pp. 440–455, Jul. 2005.

[10] J. S. Kochhar, "MULTI-HOPE: A tool for multiple floor layout problems," *Int. J. Prod. Res.*, vol. 36, no. 12, pp. 3421–3435, 1998.

[11] N. N. Nordin, Z. M. Zainuddin, S. Salim, and R. R. Ponnusamy, "Mathematical modeling and hybrid heuristic for unequal size facility layout problem," *Malaysian J. Fundam. Appl. Sci.*, vol. 5, no. 1, pp. 87–89, May 2009.

[12] H. Samarghandi, P. Taabayan, and F. F. Jahantigh, "A particle swarm optimization for the single row facility layout problem," *Comput. Ind. Eng.*, vol. 58, no. 4, pp. 529–534, May 2010.

[13] R. Şahin, K. Ertoğral, and O. Türkbey, "A simulated annealing heuristic for the dynamic layout problem with budget constraint," *Comput. Ind. Eng.*, vol. 59, no. 2, pp. 308–313, Sep. 2010.

[14] K. Y. Wong, "Applying ant system for solving unequal area facility layout problems," *Eur. J. Oper. Res.*, vol. 202, no. 3, pp. 730–746, May 2010.

[15] M. Solimanpur, P. Vrat, and R. Shankar, "An ant algorithm for the single row layout problem in flexible manufacturing systems," *Comput. Oper. Res.*, vol. 32, no. 3, pp. 583–598, Mar. 2005.

[16] S. P. Singh and V. K. Singh, "An improved heuristic approach for multi-objective facility layout problem," *Int. J. Prod. Res.*, vol. 48, no. 4, pp. 1171–1194, Jan. 2008.

[17] C. W. Chen and D. Y. Sha, "Heuristic approach for solving the multi-objective facility layout problem," *Int. J. Prod. Res.*, vol. 43, no. 21, pp. 4493–4507, Feb. 2007.

[18] S. P. Singh and V. K. Singh, "Three-level AHP-based heuristic approach for a multi-objective facility layout problem," *Int. J. Prod. Res.*, vol. 49, no. 4, pp. 1105–1125, Mar. 2010.

[19] D. E. Goldberg and J. H. Holland, "Genetic algorithms and machine learning," *Mach. Learn.*, vol. 3, nos. 2–3, pp. 95–99, 1988.

[20] G. Aiello, G. La Scalia, and M. Enea, "A multi objective genetic algorithm for the facility layout problem based upon slicing structure encoding," *Expert Syst. Appl.*, vol. 39, no. 12, pp. 10352–10358, Sep. 2012.

[21] H. Pourvaziri and B. Naderi, "A hybrid multi-population genetic algorithm for the dynamic facility layout problem," *Appl. Soft Comput.*, vol. 24, pp. 457–469, Nov. 2014.

[22] J. Iqbal and C. Guria, "Optimization of an operating domestic wastewater treatment plant using elitist non-dominated sorting genetic algorithm," *Chem. Eng. Res. Des.*, vol. 87, no. 11, pp. 1481–1496, Nov. 2009.

[23] K.-Y. Lee, M.-I. Roh, and H.-S. Jeong, "An improved genetic algorithm for multi-floor facility layout problems having inner structure walls and passages," *Comput. Oper. Res.*, vol. 32, no. 4, pp. 879–899, Apr. 2005.

[24] F. Hajabdollahi, Z. Hajabdollahi, and H. Hajabdollahi, "Soft computing based multi-objective optimization of steam cycle power plant using NSGA-II and ANN," *Appl. Soft. Comput.*, vol. 12, no. 11, pp. 3648–3655, Nov. 2012.

[25] S. S. Hosseini and M. Seifbarghy, "A novel meta-heuristic algorithm for multi-objective dynamic facility layout problem," *RAIRO Oper. Res.*, vol. 50, nos. 4–5, pp. 869–890, Aug. 2016.

[26] L. Chen, C. Yan, Y. Liao, F. Song, and Z. Jia, "A hybrid non-dominated sorting genetic algorithm and its application on multi-objective optimal design of nuclear power plant," *Ann. Nucl. Energy*, vol. 100, pp. 150–159, Feb. 2017.

[27] R. Bhattacharya and S. Bandyopadhyay, "Solving conflicting bi-objective facility location problem by NSGA II evolutionary algorithm," *Int. J. Adv. Manuf. Technol.*, vol. 51, no. 1, pp. 397–414, Apr. 2010.

[28] M.-D. Yang, Y.-P. Chen, Y.-H. Lin, Y.-F. Ho, and J.-Y. Lin, "Multiobjective optimization using nondominated sorting genetic algorithm-II for allocation of energy conservation and renewable energy facilities in a campus," *Energy Buildings*, vol. 122, pp. 120–130 Jun. 2016.

[29] M. Z. Allahyari and A. Azab, "Mathematical modeling and multi-start search simulated annealing for unequal-area facility layout problem," *Expert Syst. Appl.*, vol. 91, pp. 46–62, Jan. 2018.

[30] H. Zhang, H. Ge, R. Pan, and Y. Wu, "Multi-objective bi-level programming for the energy-aware integration of flexible job shop scheduling and multi-row layout," *Algorithms*, vol. 11, no. 12, p. 210, Dec. 2018. doi: 10.3390/a11120210.

[31] A. Kaveh, M. R. Moghaddam, and M. Khanzadi, "Efficient multi-objective optimization algorithms for construction site layout problem," *Scientia Iranica*, vol. 25, no. 4, pp. 2051–2062, Apr. 2017.

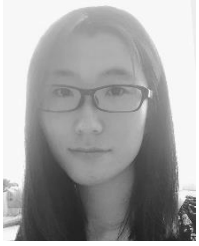
[32] B. Ulutas and A. A. Islier, "Dynamic facility layout problem in footwear industry," *J. Manuf. Syst.*, vol. 36, pp. 55–61, Jul. 2015.

[33] J. F. Gonçalves and M. G. C. Resende, "A biased random-key genetic algorithm for the unequal area facility layout problem," *Eur. J. Oper. Res.*, vol. 246, no. 1, pp. 86–107, Oct. 2015.

[34] J. Guan and G. Lin, "Hybridizing variable neighborhood search with ant colony optimization for solving the single row facility layout problem," *Eur. J. Oper. Res.*, vol. 248, no. 3, pp. 899–909, Feb. 2016.

[35] R. Matai, "Solving multi objective facility layout problem by modified simulated annealing," *Appl. Math. Comput.*, vol. 261, pp. 302–311, Jun. 2015.

[36] A. Tayal, A. Gunasekaran, S. P. Singh, R. Dubey, and T. Papadopoulos, "Formulating and solving sustainable stochastic dynamic facility layout problem: A key to sustainable operations," *Ann. Oper. Res.*, vol. 253, no. 1, pp. 621–655, Jun. 2017.
[37] B. Turanoğlu and G. Akkaya, "A new hybrid heuristic algorithm based on bacterial foraging optimization for the dynamic facility layout problem," *Expert Syst. Appl.*, vol. 98, pp. 93–104, May 2018.
[38] Y. Chen, Y. Jiang, M. I. M. Wahab, and X. Long, "The facility layout problem in non-rectangular logistics parks with split lines," *EXPERT Syst. Appl.*, vol. 42, no. 21, pp. 7768–7780, Nov. 2015.
[39] H. Zupan, N. Herakovic, J. Zerovnik, and T. Berlec, "Layout optimization of a production cell," *Int. J. Simul. Model.*, vol. 16, no. 4, pp. 603–616, Dec. 2017.
[40] G. Palubeckis, "Fast local search for single row facility layout," *Eur. J. Oper. Res.*, vol. 246, no. 3, pp. 800–814, Nov. 2015.
[41] M. Rubio-Sánchez, M. Gallego, F. Gortázar, and A. Duarte, "GRASP with path relinking for the single row facility layout problem," *Knowl. Based Syst.*, vol. 106, pp. 1–13, Aug. 2016.
[42] L. Garcia-Hernandez, H. Pierreval, L. Salas-Morera, and A. Arauzo-Azofra, "Handling qualitative aspects in unequal area facility layout problem: An interactive genetic algorithm," *Appl. Soft Comput.*, vol. 13, no. 4, pp. 1718–1727, Apr. 2013.
[43] R. Kia, F. Khaksar-Haghani, N. Javadian, and R. Tavakkoli-Moghaddam, "Solving a multi-floor layout design model of a dynamic cellular manufacturing system by an efficient genetic algorithm," *J. Manuf. Syst.*, vol. 33, no. 1, pp. 218–232, Jan. 2014.
[44] X. Wu, C.-H. Chu, Y. Wang, and W. Yan, "A genetic algorithm for cellular manufacturing design and layout," *Eur. J. Oper. Res.*, vol. 181, no. 1, pp. 156–167, Aug. 2007.
[45] A. Sadrzadeh, "A genetic algorithm with the heuristic procedure to solve the multi-line layout problem," *Comput. Ind. Eng.*, vol. 62, no. 4, pp. 1055–1064, May 2012.
[46] Z.-H. Zhang, C.-Q. Zhong, Z.-Z. Xu, and H.-F. Teng, "A non-dominated sorting cooperative co-evolutionary differential evolution algorithm for multi-objective layout optimization," *IEEE Access*, vol. 5, pp. 14468–14477, 2017. doi: 10.1109/ACCESS.2017.2716111.
[47] M. J. Mawdesley and S. H. Al-Jibouri, "Proposed genetic algorithms for construction site layout," *Eng. Appl. Artif. Intell.*, vol. 16, nos. 5–6, pp. 501–509, Aug./Sep. 2003.
[48] H. Li, R. Liu, and L. Shi, "The layout optimization problem of automobile engine production line," in *Proc. IEEE Int. Conf. Ind. Eng. Eng. Manage. (IEEM)*, Bangkok, Thailand, Dec. 2018, pp. 1736–1740.
[49] A. Azadeh, M. Moghaddam, T. Nazari, and M. Sheikhalishahi, "Optimization of facility layout design with ambiguity by an efficient fuzzy multivariate approach," *Int. J. Adv. Manuf. Technol.*, vol. 84, nos. 1–4, pp. 565–579, Apr. 2016.
[50] A. Klausnitzer and R. Lasch, "Optimal facility layout and material handling network design," *Comput. Oper. Res.*, vol. 103, pp. 237–251, Mar. 2019.



XUJING ZHANG was born in China. She received the master's degree from Soochow University, Suzhou, China, in 2013, where she is currently pursuing the degree with the College of Textile and Clothing Engineering. Her research interests include garment production management and analysis of textile comfort.



LICHUAN WANG was born in China, in 1983. He received the Ph.D. degree in computer engineering from Lille 1 University, France, in 2002. Since 2018, he has been an Associate Professor with the Textile and Clothing Engineering College, Soochow University, China. His research interests include clothing industry engineering, and textile and garment supply chain management.



YAN CHEN received the Ph.D. degree in clothing engineering from Soochow University, China, in 2003, where she is currently a Professor with the Textile and Clothing Engineering College. She is also a member of the Executive Committee of the International Joint Research and Development of the Doctoral Program SMDTex of the European Commission for Science, Technology, and Culture. Her research interests include intelligent production of clothing, analysis of textile comfort and consumer well-being, and smart textile design.